

TEN MINUTES OF DYNAMIC STRETCHING IS SUFFICIENT TO POTENTIATE VERTICAL JUMP PERFORMANCE CHARACTERISTICS

OLFA TURKI,¹ ANIS CHAOUACHI,¹ ERIC J. DRINKWATER,² MOKTAR CHTARA,¹ KARIM CHAMARI,¹ MOHAMED AMRI,³ AND DAVID G. BEHM⁴

¹Tunisian Research Laboratory "Sports Performance Optimization," National Center of Medicine and Science in Sports (CNMSS), Tunis, Tunisia; ²School of Human Movement Studies, Charles Sturt University, Bathurst, New South Wales, Australia; ³Laboratory of Functional Neurophysiology and Pathology, Faculty of Sciences, Tunis, Tunisia; and ⁴School of Human Kinetics and Recreation, Memorial University of Newfoundland, St. John's Newfoundland, Canada

ABSTRACT

Turki, O, Chaouachi, A, Drinkwater, EJ, Chtara, M, Chamari, K, Amri, M, and Behm, DG. Ten minutes of dynamic stretching is sufficient to potentiate vertical jump performance characteristics. *J Strength Cond Res* 25(9): 2453–2463, 2011—The current literature recommends dynamic rather than static stretching for the athletic warm-up. Dynamic stretching and various conditioning stimuli are used to induce potentiation in subsequent athletic performance. However, it is unknown as to which type of activity in conjunction with dynamic stretching within a warm-up provides the optimal potentiation of vertical jump performance. It was the objective of the study to examine the possible potentiating effect of various types of conditioning stimuli with dynamic stretching. Twenty athletes participated in 6 protocols. All the experimental protocols included 10 minutes of dynamic stretching. After the dynamic stretching, the subjects performed a (a) concentric (DS/CON): 3 sets of 3 repetition maximum deadlift exercise; (b) isometric (DS/ISOM): 3 sets of 3-second maximum voluntary contraction back squats; (c) plyometric (DS/PLYO): 3 sets of 3 tuck jumps; (d) eccentric (DS/ECC): 3 modified drop jumps; (e) dynamic stretching only (DS), and (f) control protocol (CON). Before the intervention and at recovery periods of 15 seconds, 4, 8, 12, 16, and 20 minutes, the participants performed 1–2 maximal countermovement jumps. The DS and DS/CON protocols generally had a 95–99% likelihood of exceeding the smallest worthwhile change for vertical jump height, peak power, velocity and force. However, the addition of the deadlift to the DS did not augment the potentiating effect. Time-to-peak potentiation was variable between individuals but was most consistent

between 3 and 5 minutes. Thus, the volume and the intensity associated with 10 minutes of dynamic stretching were sufficient to provide the potentiation of vertical jump characteristics. Additional conditioning activities may promote fatigue processes, which do not permit further potentiation.

KEY WORDS postactivation potentiation, flexibility, fatigue, plyometrics, recovery

INTRODUCTION

Power is an essential determinant of many types of athletic performance, and its optimization during training and competition can be enhanced through the use of an appropriate active warm-up (10,26,31,65). Research has indicated that the inclusion of maximal or near-maximal muscular actions may acutely improve neuromuscular performance by inducing postactivation potentiation (PAP) (74). Post-PAP performance enhancement has been associated with enhanced motor unit excitability (39), increased motor unit recruitment and synchronization, decreased presynaptic inhibition, or greater central activation of the motor neuron (1). At the level of the muscle, PAP-induced phosphorylation of the myosin light chain can also result in an increased actin-myosin crossbridge cycling rate (69).

Vertical jump performance parameters have been widely used to indirectly ascertain the occurrence of PAP in humans. Although many previous studies have reported significant improvements in vertical jump following interventions designed to induce PAP (17,37,38,56,73,88,93), others have not observed significant changes in performance (24,52,53,55,76,82). These discrepancies in outcomes are possibly because of a number of methodological PAP issues such as the volume (32), recovery period (56), and type of muscular action (dynamic, isometric) used during the potentiating exercise (24,38,45,58,63).

Address correspondence to Dr. David G. Behm, dbehm@mun.ca.

25(9)/2453–2463

Journal of Strength and Conditioning Research

© 2011 National Strength and Conditioning Association

The level of performance attained post-PAP treatment is thought to be the product of a balance between PAP and fatigue (6,83). This balance might be affected, among other things, by the rest interval between the performance testing and the PAP intervention (83). Past research has suggested recovery durations from 5 minutes (80), to 8–12 minutes (56), and 18 minutes (17) to optimize the PAP response. However, French et al. (32) and Gourgoulis et al. (37) reported that vertical jump performance is enhanced immediately post-PAP treatment. These conflicting outcomes present no clear guidelines regarding the optimal duration of recovery to maximize subsequent explosive performance.

Another factor that could affect the balance between PAP and fatigue is the type of muscle action used as a potentiating exercise. In theory, different types of contraction would have different effects on neuromuscular fatigue (2,54). Previous research investigating the occurrence of PAP in human muscles has used either isometric (7,36,38,71) or dynamic (4,16,37,41,52,56,61,73,93) actions. The isometric protocols consisted of maximum voluntary contraction (MVC) back squat (71), isometric leg-press (38), and isometric knee extensions (32). Dynamic protocols essentially consisted of dynamic back squats (17,37,41,52,61). In addition, a very limited number of investigations have used eccentric (45) and plyometric actions (63,82) as a means of enhancing subsequent vertical jump performance. Moreover, to the best of the authors' knowledge, only 4 previous protocols have directly compared different types of potentiating contractions with respect to their effect on subsequent explosive performances (24,70,82). Rixon et al. (70) used isometric and dynamic muscular actions and reported significant increases in countermovement jump (CMJ) height and peak power after an isometric condition, whereas the dynamic condition induced no change in the CMJ height but an increase in CMJ peak power. Additional research (82) investigated the PAP effects of concentric, plyometric, and isometric MVCs on sprint and jump performances and found no significant group PAP effect on sprint and jump performances. Finally, Esformes et al. (24) investigated PAP by examining the effect of heavy weight or plyometric exercises on repeated CMJ performance and compared CMJ performances between the 2 modes of exercise. The authors showed that heavy weight-induced PAP improved the CMJ jump height in comparison with plyometric exercises that presented no benefit. These results present no clear relationship between the acute performance changes and the type of muscular contraction athletes could complete during the warm-up. However, the small number of studies and the contradicting results merit further investigation regarding the appropriate type of potentiating muscular actions to be undertaken during a warm-up preceding an explosive muscular action such as the vertical jump.

Behm and Chaouachi (8) in an extensive review reported that dynamic stretching routines are more helpful than are traditional static stretching to improve explosive performance.

Although there are contradictory conclusions on whether static stretching will decrease (21,34,90) or have no significant effect (42,57,78,89) on the likelihood of musculotendinous injury, there are no review articles to our knowledge indicating that dynamic stretching will increase the chances of injury. Some studies have indicated that dynamic stretching provides similar acute increases in static flexibility as static stretching does (5,44). Dynamic stretching has been reported to either facilitate power (62,92) and jump (47,51,68) performance or have no adverse effect (18,84,86). Thus, according to Behm and Chaouachi (8), the use of dynamic stretching rather than static stretching for the warm-up would tend to be a more judicious choice. Hence, the dynamic stretching-induced facilitation of performance can be augmented by further conditioning stimuli. If greater potentiation is possible, then what is the type of contraction that is the most optimal to achieve these gains?

Only 1 previous study has investigated the effect of static stretching combined with PAP exercises on vertical jump performance (35). The authors concluded that heavy loads squatting and heavy loads squatting plus static stretching exercises had no significant effect on maximal squat jump and CMJ performances in untrained subjects. Furthermore, there are no similar studies that have recruited trained individuals despite the fact that these complex warm-up procedures are much more commonly used in this population.

In this context, no previous study has examined performance changes after a warm-up routine that includes dynamic stretching combined with different PAP protocols involving different modes (concentric, eccentric, plyometric, and isometric) of muscular actions. Therefore, the aim of this study was to determine the (a) effect of dynamic stretching combined with 4 types of PAP exercises involving different types of muscular actions on subsequent vertical jump performance in highly trained ball-game players and (b) specific optimal recovery period required for each of the 4 experimental conditions with respect to their effect on vertical jump performance parameters. It was hypothesized that the addition of the PAP exercises to dynamic stretching would enhance the potentiation of the jump performances.

METHODS

Experimental Approach to the Problem

Over a 3-week period, the participants completed 6 protocols all of which involved a standardized general warm-up, a baseline (preintervention) CMJ, and after a recovery period of 10 minutes, the participants completed 10 minutes of dynamic stretches (DS) of the lower limbs (68). After the dynamic stretch routine, the treatment protocols were completed in a counterbalanced sequence on separate days with (a) concentric protocol (DS/CON): 3 sets of 3 repetition maximum (3RM) deadlift exercise; (b) isometric protocol (DS/ISOM): 3 sets of 3-second MVC back squat; (c) plyometric protocol (DS/PLYO): 3 sets of 3 tuck jumps; (d) eccentric protocol (DS/ECC): 3 modified drop jumps (DJs); (e) dynamic stretching only protocol (DS);

and (f) the control protocol (CON). To ascertain the extent and the duration of any possible potentiation, after each condition and during recovery, 1–2 maximal CMJs were performed at 15 seconds, 4, 8, 12, 16, and 20 minutes.

Subjects

This study was conducted on 20 highly trained, male athletes (handball $n = 9$, soccer $n = 6$, and basketball $n = 5$) who also were sports science students pursuing degrees in Exercise Science and Physical Education at the University of Sports of Tunisia (Table 1). All the subjects were members of national team sports of the first division of Tunisia. The players were all starters in the competitive season when the testing was conducted. They participated in a regular training and competition schedule. They trained 5–6 times a week (~90 minutes per session) with a competitive match taking place during the weekend in their national-level championships. All the subjects had routinely performed back squats and deadlifts as part of their regular resistance training regime for a minimum of 2 years before the study. The participants were free of current or ongoing neuromuscular diseases or musculoskeletal injuries that could hinder their ability to deliver maximal effort. None of the players were taking exogenous anabolic-androgenic steroids or other drugs or substances expected to affect physical performance or hormonal balance during this study. Written informed consent was obtained from all the players after they were given a verbal and written explanation of the experimental design and the potential risks involved in the study. The study was conducted according to the Declaration of Helsinki, and the protocol was fully approved by the Clinical Research Ethics Committee and the Ethic Committee of the National Centre of Medicine and Science of Sports of Tunisia before the commencement of the assessments. All the players were fully accustomed to the procedures used in this research and were informed that they could withdraw from the study at any time without penalty.

Procedures

All the tests and measurements were conducted in an indoor facility at the university's resistance training room of the

University of Sports Science of Tunisia. One week before the commencement of the study, all the subjects participated in an orientation session to become familiar with the testing procedure. The participants were coached on proper back squat and deadlift lifting techniques using the bar. They were also instructed on proper modified drop and tuck-jump techniques and how to use the force platform for the vertical jumping test. After this familiarization session, the subjects attended the laboratory on 8 separate occasions. The objective of the first session was to determine the subjects 3RM on the back squat. During the second session, the participants completed a second strength test to determine their 3RM on the deadlift exercise. One week later, the subjects returned to the university's resistance training room on 6 separate occasions for the testing intervention sessions.

Sessions 3–8 were performed over a 3-week period, and all involved the same procedure. Each session began with the subjects performing a standardized general warm-up comprising 5 minutes of light-intensity jogging. After this, the subjects completed a baseline (preintervention) CMJ. After a recovery period of 10 minutes, the participants completed 10 minutes of DS of the lower limbs (68). The subjects then performed 1 of the PAP protocols. The subjects performed the following protocols on separate days in a counterbalanced sequence. The treatment protocols were as follows: (a) concentric protocol (DS/CON): 3 sets of 3RM deadlift exercise; (b) isometric protocol (DS/ISOM): 3 sets of 3-second MVC back squat; (c) plyometric protocol (DS/PLYO): 3 sets of 3 tuck jumps; (d) eccentric protocol (DS/ECC): 3 modified DJs; (e) dynamic stretching only protocol (DS); and (f) the control protocol (CON). Finally, after each experimental condition and during the recovery interval after the PAP protocol, the participant performed 1–2 maximal CMJs at the following time points of 15 seconds, 4, 8, 12, 16, and 20 minutes. In the DS-only and the CON conditions, the subject was required to perform the 6 recovery session blocks of maximal CMJs immediately after the dynamic stretching protocol, and after the 5 minutes of light-intensity jogging warm-up, respectively. Each experimental protocol was applied in a counterbalanced, randomized order on separate days. Testing took place on the same time of the day for each subject and with a minimum of 72 hours between testing sessions. The participants were instructed to maintain their usual weekly training routines during the course of the study. The subjects refrained from any strenuous activities, resistance, or plyometric training at least 48 hours before each testing session.

To minimize confounding factors, instructions related to sleep and diet were given to the subjects before the experiment. On the night preceding each test session, the subjects were asked to keep their usual sleeping habits, with a minimum of 6 hours of sleep. During the period of investigation, they were prohibited from consuming any known stimuli (e.g., caffeine) or depressants (e.g., alcohol) that

TABLE 1. Subject characteristics ($N = 20$).*

Variable	Mean $\pm$ SD
Age (y)	20.7 $\pm$ 1.3
Height (m)	178.9 $\pm$ 5.4
Body mass (kg)	73.4 $\pm$ 6.9
Body fat (%)	10.8 $\pm$ 2.5
Back squat 3RM (kg)	141.9 $\pm$ 19.7
Deadlift 3RM (kg)	131.62 $\pm$ 17.53

*RM = repetition maximum.

could possibly enhance or compromise wakefulness. Moreover, the participants were requested to maintain their habitual physical activity and to avoid strenuous activity in the day before and throughout the study. To avoid dehydration, ad libitum drinking was permitted for the athletes during all the testing sessions.

Three Repetition Maximum Procedures

The subjects' 3RM back squat and deadlift exercises were tested using the procedure outlined by the National Strength and Conditioning Association (3). The 3RM was defined as a load that caused failure on the fourth repetition without the loss of proper exercise technique. Based on this first testing day, the value of 1RM was estimated for each subject, according to the percentage 1RM-repetition relationship outlined by Baechle et al. (3). The resistance for the isometric conditioning stimuli of the isometric protocol was then calculated from each subject's 1RM load. Before the start of the strength testing sessions, all the subjects underwent a standardized general warm-up that comprised light-intensity jogging for 5 minutes, followed by a series of dynamic movements with an emphasis on warming up the musculature associated with the lift. The subjects then performed a weights-specific warm-up involving 8 repetitions at 50% 1RM, 4 repetitions at 70% 1RM, and finally 2 repetitions at 80% of 1RM, with 3-minute intervals between them. During the course of the 2 strength testing sessions, the subjects used their estimated 1RM back squat and deadlift exercises as a guide. After the final warm-up set, the subjects attempted 3 repetitions of a set load (3RM), and if successful, the resistance was increased until the subject could not lift the weight through the full range of motion. A 5-minute rest was imposed between all the attempts to allow the subjects to recover. The 3RM was determined after 3–4 attempts in all the subjects. During the squat 3RM test, each subject was required to descend to the "parallel" position where the greater trochanter of the femur was aligned with the knee (a line between the lateral epicondyle of the femur and the greater trochanter was approximately parallel to the floor) and ascended until full knee and hip extension. For the deadlift 3RM test, the subject's feet and hands were spaced evenly from the center of the bar and were allowed to be placed close to its center (power style). The knees were flexed at 100° (19). When performing the lift, the subjects kept the back rigid and arms straight to lift the bar using the legs and hips while keeping the bar as close to the body as possible. The participant lifted the bar vertically from the floor with 1 smooth motion until the knees and back extended the body to an erect position and could not bounce the weight off the floor. The investigator (certified strength and conditioning specialist) was located lateral to the subject and gave a verbal signal once the athlete had reached the appropriate depth and when the knee became fully extended, respectively, for the squat and deadlift exercises. The proper technique was ensured by the investigator and verbal feedback provided to

the subjects that they had reached the required depth for the squat and the necessary extension for the deadlift exercise. A complete range of motion and proper technique were required for each successful 3RM trial.

The reliability of the 3RM squat and deadlift was tested over a 2-day period before experimentation. Excellent intraclass correlation coefficients of 0.98 and 0.97 were achieved for the squat and deadlift, respectively.

Stretching Protocol

The dynamic stretching protocol was chosen after an analysis of the warm-up practices performed by the subjects before typical training sessions and should be considered as representing their self-selected, preferred warm-up modality. The 10-minute duration dynamic stretch protocol incorporated 5 active dynamic exercises to dynamically stretch the lower-body musculature mainly used in jumping (gastrocnemius, gluteals, hamstrings, quadriceps, and hip extensors). The list, order, and description of the dynamic stretches can be found in Table 2. All the exercises were performed while walking over a distance of 20 m. The movements were carried out approximately 14 times for each exercise. Using a randomized selection procedure; the participants underwent 4 sets of dynamic stretches over a 20-m distance on each leg independently. A rest period of 10 seconds was allowed between sets before returning to the start position. The participants were continually instructed to maintain a vertical torso, with knees toward chest, while performing the active dynamic stretching exercises. The active dynamic stretches have been used previously (68,85).

Protocols for Eliciting Postactivation Potentiation

The DS/CON protocol required the subjects to perform 3 repetitions of the dynamic deadlift at an intensity of 3RM, with a resting time of 4 minutes between conditions. For the DS/ISOM protocol, the participants underwent 3 repetition of 3-second MVC of the isometric squat (32,70). The DS/PLYO protocol involved 3 sets of 3 maximal repetitions of the double-legged tuck-jump exercise. The double-leg tuck jump is a medium-intensity plyometric drill (63) and is performed by explosively jumping upward while quickly pulling the knees to the chest. The subjects were instructed to perform to the best of their ability and to minimize ground contact time and maintain the correct technique on each repetition. The DS/ECC protocol consisted of 3 modified DJs from a height of 80 cm. The modified DJs were standardized to the departure position (0° knee flexion) and landing position (active stabilization in 90° knee flexion), which are reported to provoke high eccentric muscle action (45).

Jump Testing

Vertical jump performance was assessed on Quattro Jump portable force plate (Kisler Instrument AG, Winterthur, Switzerland) at a sampling rate of 500 Hz. The participants performed a CMJ according to the protocol described by Bosco et al (11). The subjects were asked to keep their hands

TABLE 2. Dynamic stretching exercises.

Muscle groups	Dynamic stretching exercises
Gluteals	Walking high bringing the knee to the chest. While walking, lift the knee toward the chest, raise the body on the toes of the opposite extended leg
Hamstrings	Walking while swinging actively the leg to be stretched forward into hip flexion until a stretch is felt in the posterior thigh while keeping the knee extended and the ankle in plantar flexion
Adductors	Hurdler's knee raises forward movement. While traveling forward, the participant raises the trailing leg and places the hip in flexion ($\sim 90^\circ$) in an abducted and externally rotated position, with the knee flexed at 90° . In this position, the limb is displaced forward as though the participants were stepping over an object just below waist height and returned to normal walking stride position
Quadriceps	Heel ups. Rapidly kick heels toward buttocks while moving forward
Gastrocnemius	Tip-toe walking. Traveling forward while completing alternating plantar flexion (tip-toe) with every step forward. The aim is to raise the body as high as possible through tip-toeing

on their hips (akimbo) throughout the entire jump to minimize lateral and horizontal displacement during performance, to prevent any influence of arm movements on the vertical jumps, and to avoid coordination as a confounding variable in the assessment of the leg extensors neuromuscular performance (13). The players were asked to jump as high as possible, and the greatest jump height and associated maximal velocity before take-off, maximal force before the take-off, and peak power were used for analysis. Jump height was determined as the center of mass displacement, calculated from the recorded force and body mass. The reliability of the CMJ test has been shown in previous studies to range from 0.95 to 0.96 (13,14).

Statistical Analyses

The peak of each dependent variable was compared with the baseline, regardless of at which time point the peak occurred. Within-subject contrasts were conducted from a repeated-measure analysis of variance (ANOVA) between the baseline and peak for each dependent variable. Repeated-measures ANOVAs were calculated using Predictive Analytics Software (version 17.0.2). From the resulting p value, mean difference between comparisons, and total SD of the baseline and peak data, Cohen's d , smallest worthwhile change (SWC), and likelihood of clinical meaningfulness were calculated (59). The Cohen's d is calculated from the mean change divided by the SD of the data; thresholds for qualitative descriptors of Cohen's d were set at <0.20 is "trivial," 0.20 – 0.49 is "small," 0.50 – 0.79 is "moderate," and >0.80 is "large" (20). The smallest change to be considered worthwhile (SWC) was thus calculated from 0.20 of the SD of the data. The threshold of a clinical meaningful effect was

set at 75% (59). Descriptive statistics, p values, 95% confidence limits, and Cohen's d for the within-subject contrasts were calculated by custom-written Excel spreadsheets (Microsoft Office, 2007). Magnitude-based inferences were then calculated using a second custom-written Excel spreadsheet (48).

RESULTS

Null-Hypothesis Significance Testing

Time. The only main effect for time was the decrement in most dependent variables by 8 minutes. The CMJ height, power, and velocity had declined by 1.1–3.0, 1.8–3.7, and 0.5–1.7%, respectively, between 8 and 20 minutes.

Activity. There were no statistically significant differences between groups in velocity ($p = 0.074$), power ($p = 0.058$), or force ($p = 0.698$). There were statistically significant differences between control and DS/CON (2.9%, $p = 0.016$) and control and DS (2.9%, $p = 0.001$) in the CMJ height.

Countermovement Jump Height

The 2 warm-up conditions that elicited a substantial likelihood of potentiating CMJ height a substantial amount (i.e., had a $>75\%$ of exceeding a small Cohen's d) were the DS/CON and DS conditions (Table 3).

Power

Although the DS/ECC and DS warm-up conditions both exceeded the 75% likelihood of eliciting a meaningful potentiation of power, it should also be noted that the DS/CON warm-up also approached our 75% threshold (Table 3).

TABLE 3. The precision of the predicted increase from baseline to peak of each warm-up protocol on each of the dependent variables.*†

Condition	Mean difference	95% Confidence limits		Cohen's <i>d</i>	Likelihood of exceeding smallest worthwhile change (%)		No. of subjects whose peak was at baseline (total <i>n</i> = 21)	Time to peak (min)
		Lower	Upper		Higher	Trivial		Mean (<i>SD</i>)
Vertical jump height (cm)								
Control	0.39	−0.20	0.98	0.08	3	97	9	7.6 (8.1)
Plyo	0.50	−0.39	1.38	0.10	13	87	11	10.9 (8.1)
Iso	0.39	−0.15	0.93	0.09	5	95	9	4.9 (3.1)
Conc	2.71	1.54	3.89	0.67	>99	0	1	4.0 (5.2)
Ecc	0.77	0.04	1.51	0.17	37	63	9	8.9 (7.0)
DS	2.20	1.56	2.85	0.53	>99	0	0	5.3 (6.1)
Peak power (W)								
Control	0.43	0.12	0.74	0.09	0	>99	9	5.0 (6.0)
Plyo	0.94	0.28	1.61	0.20	52	48	12	12.1 (7.9)
Iso	0.98	0.40	1.55	0.18	38	62	9	3.6 (3.5)
Conc	1.19	0.52	1.85	0.24	72	28	8	9.7 (7.2)
Ecc	1.44	0.98	1.89	0.32	99	1	3	5.0 (5.1)
DS	1.60	1.04	2.17	0.38	99	1	2	3.9 (4.7)
Velocity (m·s ^{−1})								
Control	0.02	0.01	0.03	0.10	0	>99	10	6.8 (7.4)
Plyo	0.04	0.02	0.06	0.25	76	24	5	4.3 (4.4)
Iso	0.03	0.02	0.04	0.19	44	56	6	3.2 (2.0)
Conc	0.04	0.03	0.05	0.29	95	5	1	4.7 (5.5)
Ecc	0.02	0.00	0.04	0.15	19	81	9	8.6 (7.8)
DS	0.05	0.03	0.06	0.31	97	3	3	3.1 (4.5)
Force (N)								
Control	77.26	37.52	116.99	0.34	94	6	9	7.9 (6.2)
Plyo	99.96	39.04	160.89	0.48	97	3	6	7.9 (6.4)
Iso	122.50	72.32	172.68	0.51	>99	0	4	7.2 (7.7)
Conc	114.37	60.32	168.43	0.50	99	1	4	9.5 (5.8)
Ecc	159.63	99.10	220.15	0.64	>99	0	2	9.5 (6.4)
DS	105.28	59.82	150.73	0.44	99	1	5	7.6 (5.7)

*DS = dynamic stretching; Plyo = plyometric; Ecc = eccentric; Conc = concentric; Iso = isometric.

†Also illustrated are the standardized effect size and the percent likelihood that the predicted effect will exceed the smallest worthwhile effect. Many participants had as their overall peak for the testing session the baseline measure, thus never reaching any "potentiation" at all. When the peak achieved was not the baseline, the substantial variability in the time to the dependent variable peaking after baseline measurements is also illustrated.

Velocity

The DS/PLYO and DS/CON warm-up conditions, in addition to the DS conditions, each illustrated a substantial likelihood of potentiating jump velocity (Table 3).

Peak Force

Force generated during the vertical jump was potentiated for all the conditions, including the control condition (Table 3).

DISCUSSION

The most important findings of this study were that the DS protocol consistently had a 97–99% likelihood of potentiating the vertical jump height, peak power, velocity, and force by at

least the SWC. The addition of a concentric deadlift exercise to the DS (DS/CON) provided 95, 99, and 72% likelihoods of exceeding the SWC for the vertical jump height, velocity, and peak power, respectively.

The results are presented in terms of magnitude-based inferences rather than null-hypothesis significance testing to distinguish between the probability that a difference between the 2 conditions is zero (i.e., null-hypothesis significance testing) vs. the likelihood that a difference may be meaningful in an athletic setting (i.e., magnitude-based inferences). For example, consider that although the improvement in peak power of the control group had a very low probability of actually being zero ($p < 0.05$), there is little practical

application for an improvement of $<0.5 \text{ W} \cdot \text{kg}^{-1}$. What value is there in testing if the change was zero? Of much greater interest is if the intervention will have a *meaningful* effect. With an effect size of 0.09, the potentiation of peak power is almost certain to have little practical value.

Additionally, we opted to assess the peak of each dependent variable in each condition from each subject, regardless of the time point at which it occurred. When using conventional null-hypothesis significance testing to assess for differences between time points, not only were there not any signs of potentiation but most variables (CMJ height, power, and velocity) were significantly *decreased* from 8 minutes while force remained significantly decreased from 12 minutes. Furthermore, although "... God surely loves the 0.06 nearly as much as the 0.05," (72) the strict interpretation of statistical significance indicates that there were no statistically significant differences between groups in velocity ($p = 0.074$) nor power ($p = 0.058$). However, our results presented in Table 3 demonstrate that there are certain warm-up activities that are distinctly more likely to elicit substantial potentiation than others. Note that making magnitude-based inferences is not a more liberal method of interpreting data but is about describing the likelihood that there may be a useful change in performance (23). Although we are suggesting that DS and concentric-type warm-up activities are likely to induce potentiation, it is based on the results presented in Table 3, not on those of the statistically significant findings of the CMJ. Although some changes were statistically significant, they were too small to be clinically meaningful, whereas others may not have been statistically significant but have a substantial likelihood of making a substantial contribution to performance. If the change in the dependent variable is zero, it is not a relevant question; if the change may have a meaningful effect on performance, it is a much more relevant way to phrase the research question.

The potentiation with dynamic stretching in this study is not without precedent. The literature indicates that dynamic stretching can show facilitation of power (62,91,92), sprint (28,60,85), and jump (47,49,68) performance. There are also studies that report no effect of dynamic stretching on performance (14,18,75,84,86). Behm and Chaouachi (8) in an extensive review reported that shorter durations of dynamic stretching do not typically adversely affect performance, whereas longer duration of dynamic stretches generally facilitate performance (33,49,68,77,92). In line with the analysis of Behm and Chaouachi's review (8), the DS protocol in this study used 10 minutes of dynamic stretching. For instance, 10 minutes of dynamic warm-up activities (stretching or aerobic activity) have been reported to result in improvements in shuttle run time, medicine ball throw distance, and 5-step jump distance (65) and a tendency ($p = 0.06$) for increased jump height (22). Hough et al. (49) instituted 7 minutes of dynamic stretching resulting in an increased vertical jump height and electromyographic (EMG) activity.

The mechanisms by which dynamic stretching improves muscular performance have been suggested to be elevated heart rate and muscle and body temperature (29,30,91). The voluntary contractions associated with dynamic stretching have been reported to enhance motor unit excitability and improve kinesthetic sense leading to improved proprioception and preactivation (8) improved musculotendinous unit stiffness and increase nerve impulse transmission leading to favorable changes in the force-velocity relationship (10) and the decreased inhibition of antagonist muscles (51,91). As a result of these effects, dynamic stretching may enhance force and power development (49,84,91). Indeed, Faigenbaum et al. (25) and Yamaguchi and Ishii (91) hypothesized that the increases in force output after dynamic stretching are caused by an enhancement of neuromuscular function, and they implied that the dynamic stretching had a PAP effect on performance via an increase in the rate of crossbridge attachments (50). However, Herda et al. (43) reported that dynamic stretching did not improve muscular strength, although EMG amplitude increased, which may alternatively reflect a potentiating effect of the dynamic stretching on muscle activation.

A major objective of this study was to determine if there was a type of activity or contraction that in conjunction with dynamic stretching might further potentiate performance. Although the DS/CON condition (dynamic stretching and deadlift) did provide 95, 99, 99, and 72% likelihood of exceeding the SWC for vertical jump height, velocity, force, and peak power, respectively, the mean differences from the DS protocol were not substantial. The DS/CON and DS protocols similarly improved vertical jump (2.7 vs. 2.1 cm; 99% likelihoods), peak power (1.2 vs. 1.6 $\text{W} \cdot \text{kg}^{-1}$; 72% vs. 99% likelihoods), velocity (0.03 vs. 0.03 $\text{m} \cdot \text{s}^{-1}$; 95 vs. 97% likelihoods), and force (113.3 vs. 105.3 N; 99% likelihoods). Although a number of studies that report subsequent vertical jump potentiation have employed maximum or near maximum intensity contractions (24,38,64,66,70,93), there are also a variety of studies inducing jump potentiation with lower intensity contractions (12,15,24,38,70,79). Chaouachi et al. (unpublished data, 2011) investigated a variety of conditioning stimuli (3–8 repetitions using 70–90% of 1RM squats) to induce the potentiation of vertical jump variables. The protocols involving submaximal intensity workloads of 5 repetitions at 70% of 1RM and 3 repetitions at 85% of 1RM had the most consistent substantial likelihood of increasing the peak of most dependent variables. The dynamic stretching routine in this study included 4 sets of dynamic stretches over a 20-m distance on each leg independently. Although the contractions were not maximal, the participants considered them vigorous.

The findings of this study would indicate that there was a workload saturation of the potentiating effect. In addition to the dynamic stretching, the increased deadlift workload did not augment the potentiation or in other words there was a plateau in the dose-response relationship. Our findings

contradict those of Needham et al. (67) who reported that 10 minutes of dynamic warm-up with the inclusion of 8 front squats at 20% of the body mass enhanced jumping ability more than dynamic exercise alone did. In line with these observations, it has been reported that continually wearing a weighted vest for the entire (2% of body mass) or for the last 4 exercises (10% of body mass) of a dynamic warm-up improved vertical jump (46), and long jump (81) performances more than a warm-up consisting of only dynamic stretching. The disparity in the conclusions between our study and that of the aforementioned studies may be because of the differing intensities of the potentiating protocol that was added to dynamic stretching during the warm-up process.

There is considerable evidence that activity-dependent potentiation results from the balance between PAP and fatigue (6,7,69,83). This balance might be affected, among other things by the volume of the conditioning contraction (7,40,87). After a relatively low conditioning contraction volume, PAP may predominate over fatigue processes and may therefore be used immediately. In contrast, as the conditioning contraction volume increases, so does fatigue. Turki et al. (85) reported that performing 1 and 2 sets of dynamic stretching over 20 m led to a significant improvement in the 20-m sprint time but that 3 sets adversely affected 20-m sprint performance, which was attributed to the presence of concurrent fatigue. The additional volume of the deadlift protocol with the dynamic stretching in this study did not provide further benefit.

The time to potentiation was quite variable between individuals in this study (Table 2: time to peak). Table 2 illustrates that the post-warm-up recovery time of likely exceeding the SWC (>95%) was typically between 3 and 5 minutes. A number of studies have commented on the extent of individual variability in muscle potentiation studies (9,80,82). In the aforementioned Chaouachi et al. (unpublished data, 2011) study (conditioning stimuli of 3–8 repetitions using 70–90% of 1RM squats), most dependent variables peaked at 1, 3, or 5 minutes though this peak was often not greater than the SWC. There were considerable interindividual variations in the potentiation of power and velocity. Other studies have encountered similar results. McCann and Flanagan (64) illustrated the individual variability in vertical jump height potentiation in their study. Although there were no significant effects reported by Till and Cooke (82) for any PAP treatment on sprint and vertical jump measures, there were substantial improvements made by the majority of subjects. Thus, although this study and others indicate the greatest potentiating effect up to approximately 5 minutes poststimulus, the variability encountered in these studies emphasizes the importance of examining individual training routines.

Another interesting finding was that force was potentiated by all the conditions including the control condition and that the mean time to peak force potentiation was 7–10 minutes.

Although the mean difference (pre vs. post-warm-up) and effect size were the least for the control condition, a warm-up involving a general jog and run for 5 minutes and a single baseline CMJ provide a 94% likelihood of exceeding the SWC. The control condition did not potentiate any other test measures. Thus, although force potentiation was more sensitive to prior activity in this study, it was not directly related to vertical jump height enhancement. Flanagan and McCann (27) suggested that jump height may be more related to force-velocity relations, increased use of elastic energy, and improved rate of force development rather than peak force alone. Chaouachi et al. (unpublished data, 2011) in their potentiation study found trivial increases in vertical jump height that contrasted with power and velocity measures, which exceeded a 75% likelihood of providing a worthwhile change. The dichotomy was explained by indicating that a CMJ involves not only power and velocity but also the coordination and sequencing of the power and velocity of each limb segment. If segmental coordination is altered, there may not be an appropriate summation of forces leading to no improvement of vertical jump height. Thus, in this study, an increase in force achieved by the control, DS/PLYO, DS/ISO, and DS/ECC conditions did not sufficiently augment the other aforementioned integrative factors involved with increased vertical jump height.

In conclusion, 10 minutes of dynamic stretching of the lower limbs has a substantial likelihood of augmenting vertical jump height, peak power, velocity, and force. The addition of a concentric deadlift exercise to the DS (DS/CON) also exceeded the SWC for vertical jump height, velocity, and peak power, respectively. However, because it was unlikely that there was any additional benefit of adding the deadlift exercises to the DS, the increased volume of the warm-up intervention did not provide substantial further benefit. These findings may be indicative of a potentiating dose-response relationship between PAP and fatigue processes. As in other studies, the time-to-peak potentiation was variable between individuals; however, the most likely time of exceeding the SWC ranged between 3 and 5 minutes postintervention.

PRACTICAL APPLICATIONS

It is possible that to enhance subsequent explosive performance, moderate-intensity conditioning contractions can be more appropriate than that at high intensities when added to 10 minutes of dynamic stretching during the warm-up. Because dynamic stretching has been shown to increase static and dynamic range of motion (5,44), optimizing explosive type athletic performances should involve vigorous dynamic stretching activities of moderate duration (5–10 minutes) (8). Because the potentiation of a prior conditioning program is very individualized, the timing of warm-up activities needs to be developed individually for each athlete. However, when individual warm-ups are not possible as with a team setting, then a 3- to 5-minute recovery period is generally the most appropriate. Future research should investigate the effect of

different intensities of conditioning contraction to be added to dynamic stretching to maximize explosive performance.

REFERENCES

1. Aagaard, P, Simonsen, EB, Andersen, JL, Magnusson, P, and Dyhre-Poulsen, P. Neural adaptation to resistance training: Changes in evoked V-wave and H-reflex responses. *J Appl Physiol* 92: 2309–2318, 2002.
2. Brabault, N, Desbrosses, K, Fabre, MS, Michaut, A, and Pousson, M. Neuromuscular fatigue development during maximal concentric and isometric knee extensions. *J Appl Physiol* 100: 780–785, 2006.
3. Baechle, TR, Earle, RW, and Wathen, D. *Essentials of Strength Training and Conditioning*. Champaign, IL: Human Kinetics, 2008. pp. 381–412.
4. Batista, MA, Ugrinowitsch, C, Roschel, H, Lotufo, R, Ricard, MD, and Tricoli, VA. Intermittent exercise as a conditioning activity to induce postactivation potentiation. *J Strength Cond Res* 21: 837–840, 2007.
5. Beedle, BB and Mann, CL. A comparison of two warm-ups on joint range of motion. *J Strength Cond Res* 21: 776–779, 2007.
6. Behm, DG. Force maintenance with submaximal fatiguing contractions. *Can J Appl Physiol* 29: 274–290, 2004.
7. Behm, DG, Button, DC, Barbour, G, Butt, JC, and Young, WB. Conflicting effects of fatigue and potentiation on voluntary force. *J Strength Cond Res* 18: 365–372, 2004.
8. Behm, DG and Chaouachi, A. A review of the acute effects of static and dynamic stretching on performance. *Eur J Appl Physiol* 4–3: 2011.
9. Bevan, HR, Cunningham, DJ, Tooley, EP, Owen, NJ, Cook, CJ, and Kilduff, LP. Influence of postactivation potentiation on sprinting performance in professional rugby players. *J Strength Cond Res* 24: 701–705, 2010.
10. Bishop, D. Warm up I: Potential mechanisms and the effects of passive warm up on exercise performance. *Sports Med* 33: 439–454, 2003.
11. Bosco, C, Luhtanen, P, and Komi, PV. A simple method for measurement of mechanical power in jumping. *Eur J Appl Physiol Occup Physiol* 50: 273–282, 1983.
12. Burkett, LN, Phillips, WT, and Ziuraitis, J. The best warm-up for the vertical jump in college-age athletic men. *J Strength Condition Res* 19: 673–676, 2005.
13. Chaouachi, A, Brughelli, M, Chamari, K, Levin, GT, Ben, AN, Laurencelle, L, and Castagna, C. Lower limb maximal dynamic strength and agility determinants in elite basketball players. *J Strength Cond Res* 23: 1570–1577, 2009.
14. Chaouachi, A, Castagna, C, Chtara, M, Brughelli, M, Turki, O, Galy, O, Chamari, K, and Behm, DG. Effect of warm-ups involving static or dynamic stretching on agility, sprinting, and jumping performance in trained individuals. *J Strength Cond Res* 24: 2001–2011.
15. Chattong, C, Brown, LE, Coburn, JW, and Noffal, GJ. Effect of a dynamic loaded warm-up on vertical jump performance. *J Strength Cond Res* 24: 1751–1754, 2010.
16. Chatzopoulos, DE, Michailidis, CJ, Giannakos, AK, Alexiou, KC, Patikas, DA, Antonopoulos, CB, and Kotzamanidis, CM. Post-activation potentiation effect after heavy resistance exercise on running speed. *J Strength Condition Res* 21: 1278–1281, 2007.
17. Chiu, LZ, Fry, AC, Weiss, LW, Schilling, BK, Brown, LE, and Smith, SL. Postactivation potentiation response in athletic and recreationally trained individuals. *J Strength Cond Res* 17: 671–677, 2003.
18. Christensen, BK and Nordstrom, BJ. The effects of proprioceptive neuromuscular facilitation and dynamic stretching techniques on vertical jump performance. *J Strength Cond Res* 22: 1826–1831, 2008.
19. Chulvi-Medrano, I, Garcia-Masso, X, Colado, JC, Pablos, C, de Moraes, JA, and Fuster, MA. Deadlift muscle force and activation under stable and unstable conditions. *J Strength Cond Res* 24: 2723–2730, 2010.
20. Cohen, J. *Statistical Power Analysis for the Behavioural Sciences*. Hillsdale, NJ: L. Erlbaum Associates, 1988.
21. Cross, K and Worrell, T. Effects of a static stretching program on the incidence of lower extremity musculotendinous strains. *J Athl Training* 34: 11–14, 1999.
22. Curry, BS, Chengkalath, D, Crouch, GJ, Romance, M, and Manns, PJ. Acute effects of dynamic stretching, static stretching, and light aerobic activity on muscular performance in women. *J Strength Cond Res* 23: 1811–1819, 2009.
23. Drinkwater, EJ. Application of confidence limits and effect sizes in sport research. *Open Sports Sci J* 1: 3–4, 2008.
24. Esformes, JL, Cameron, N, and Bampouras, TM. Postactivation potentiation following different modes of exercise. *J Strength Cond Res* 24: 1911–1916, 2010.
25. Faigenbaum, AD, Bellucci, M, Bernieri, A, Bakker, B, and Hoorens, K. Acute effects of different warm-up protocols on fitness performance in children. *J Strength Cond Res* 19: 376–381, 2005.
26. Faigenbaum, AD, McFarland, JE, Schwerdtman, JA, Ratamess, NA, Kang, J, and Hoffman, JR. Dynamic warm-up protocols, with and without a weighted vest, and fitness performance in high school female athletes. *J Athl Train* 41: 357–363, 2006.
27. Flanagan, SP and McCann, MR. Post activation potentiation is not due to increased peak ground reaction force. [Conference abstracts]. *J Strength Condition Res* 22: 2–3, 2008.
28. Fletcher, IM and Anness, R. The acute effects of combined static and dynamic stretch protocols on fifty-meter sprint performance in track-and-field athletes. *J Strength Cond Res* 21: 784–787, 2007.
29. Fletcher, IM and Jones, B. The effect of different warm-up stretch protocols on 20-meter sprint performance in trained rugby union players. *J Strength Cond Res* 18: 885–888, 2004.
30. Fletcher, IM and Monte-Colombo, MM. An investigation into the possible physiological mechanisms associated with changes in performance related to acute responses to different preactivity stretch modalities. *Appl Physiol Nutr Metab* 35: 27–34, 2010.
31. Fradkin, AJ, Zazryn, TR, and Smoliga, JM. Effects of warming-up on physical performance: A systematic review with meta-analysis. *J Strength Cond Res* 24: 140–148, 2010.
32. French, DN, Kraemer, WJ, and Cooke, CB. Changes in dynamic exercise performance following a sequence of preconditioning isometric muscle actions. *J Strength Cond Res* 17: 678–685, 2003.
33. Gelen, E. Acute effects of different warm-up methods on sprint, slalom dribbling, and penalty kick performance in soccer players. *J Strength Cond Res* 24: 950–956, 2010.
34. Gleim, G and McHugh, MP. Flexibility and its effects on sports injury and performance. *Sports Med* 24: 289–297, 1997.
35. Gonzalez-Rave, JM, Machado, L, Navarro-Valdivielso, F, and Vilas-Boas, JP. Acute effects of heavy-load exercises, stretching exercises, and heavy-load plus stretching exercises on squat jump and countermovement jump performance. *J Strength Cond Res* 23: 472–479, 2009.
36. Gossen, ER and Sale, DG. Effect of postactivation potentiation on dynamic knee extension performance. *Eur J Appl Physiol* 83: 524–530, 2000.
37. Gourgoulis, V, Aggeloussis, N, Kasimatis, P, Mavromatis, G, and Garas, A. Effect of a submaximal half-squats warm-up program on vertical jumping ability. *J Strength Cond Res* 17: 342–344, 2003.
38. Gullich, A and Schmidtbleicher, D. MVC-induced short-term potentiation of explosive force. *New Stud Athl* 11: 67–81, 1996.
39. Hamada, T, Sale, DG, MacDougall, D, and Tarnopolsky, MA. Postactivation potentiation, fiber type, and twitch contraction time in human knee extensor muscles. *J Appl Physiol* 88: 2131–2137, 2000.
40. Hamada, T, Sale, DG, MacDougall, JD, and Tarnopolsky, M. Interaction of fibre type, potentiation and fatigue in human knee extensor muscles. *Acta Physiol Scand* 178: 165–174, 2003.

41. Hanson, ED, Leigh, S, and Mynark, RG. Acute effects of heavy-and light-load squat exercise on the kinetic measures of vertical jumping. *J Strength Cond Res* 21: 1012–1017, 2007.
42. Herbert, RD and Gabriel, M. Effects of stretching before and after exercising on muscle soreness and risk of injury: Systematic review. *BMJ* 325: 468, 2002.
43. Herda, TJ, Cramer, JT, Ryan, ED, McHugh, MP, and Stout, JR. Acute effects of static versus dynamic stretching on isometric peak torque, electromyography, and mechanomyography of the biceps femoris muscle. *J Strength Cond Res* 22: 809–817, 2008.
44. Herman, SL and Smith, DT. Four-week dynamic stretching warm-up intervention elicits longer-term performance benefits. *J Strength Cond Res* 22: 1286–1297, 2008.
45. Hilfiker, R, Hubner, K, Lorenz, T, and Marti, B. Effects of drop jumps added to the warm-up of elite sport athletes with a high capacity for explosive force development. *J Strength Cond Res* 21: 550–555, 2007.
46. Hoffman, JR, Ratamess, NA, Cooper, JJ, Kang, J, Chilakos, A, and Faigenbaum, AD. Comparison of loaded and unloaded jump squat training on strength/power performance in college football players. *J Strength Cond Res* 19: 810–815, 2005.
47. Holt, BW and Lambourne, K. The impact of different warm-up protocols on vertical jump performance in male collegiate athletes. *J Strength Cond Res* 22: 226–229, 2008.
48. Hopkins, WG. A spreadsheet for deriving confidence interval, mechanistic inference and clinical inference from a p value. *Sports Science* 11: 16–20, 2007.
49. Hough, PA, Ross, EZ, and Howatson, G. Effects of dynamic and static stretching on vertical jump performance and electromyographic activity. *J Strength Cond Res* 23: 507–512, 2009.
50. Houston, ME and Grange, RW. Myosin phosphorylation, twitch potentiation, and fatigue in human skeletal muscle. *Can J Physiol Pharm* 68: 908–913, 1990.
51. Jagers, JR, Swank, AM, Frost, KL, and Lee, CD. The acute effects of dynamic and ballistic stretching on vertical jump height, force, and power. *J Strength Cond Res* 22: 1844–1849, 2008.
52. Jensen, RL and Ebben, WP. Kinetic analysis of complex training rest interval effect on vertical jump performance. *J Strength Condition Res* 17: 345–349, 2003.
53. Jones, P and Lees, A. A biomechanical analysis of the acute effects of complex training using lower limb exercises. *J Strength Cond Res* 17: 694–700, 2003.
54. Kay, D, St Clair, GA, Mitchell, MJ, Lambert, MI, and Noakes, TD. Different neuromuscular recruitment patterns during eccentric, concentric and isometric contractions. *J Electromyogr Kinesiol* 10: 425–431, 2000.
55. Khamoui, AV, Brown, LE, Coburn, JW, Judelson, DA, Uribe, BP, Nguyen, D, Tran, T, Eurich, AD, and Noffal, GJ. Effect of potentiating exercise volume on vertical jump parameters in recreationally trained men. *J Strength Cond Res* 23: 1465–1469, 2009.
56. Kilduff, LP, Bevan, HR, Kingsley, MIC, Owen, NJ, Bennett, MA, Bunce, PJ, Hore, AM, Maw, JR, and Cunningham, DJ. Postactivation potentiation in professional rugby players: Optimal recovery. *J Strength Cond Res* 21: 1134–1138, 2007.
57. Knudson, D. Stretching during warm-up: Do we have enough evidence? *JOPERD* 70: 24–28, 1999.
58. Koch, AJ, O'Bryant, HS, Stone, ME, Sanborn, K, Proulx, C, Hrubby, J, Shannonhouse, E, Boros, R, and Stone, MH. Effect of warm-up on the standing broad jump in trained and untrained men and women. *J Strength Cond Res* 17: 710–714, 2003.
59. Liow, DK and Hopkins, WG. Velocity specificity of weight training for kayak sprint performance. *Med Sci Sports Exerc* 35: 1232–1237, 2003.
60. Little, T and Williams, AG. Effects of differential stretching protocols during warm-ups on high-speed motor capacities in professional soccer players. *J Strength Cond Res* 20: 203–207, 2006.
61. Mangus, BC, Takahashi, M, Mercer, JA, Holcomb, WR, McWhorter, JW, and Sanchez, R. Investigation of vertical jump performance after completing heavy squat exercises. *J Strength Cond Res* 20: 597–600, 2006.
62. Manoel, ME, Harris-Love, MO, Danoff, JV, and Miller, TA. Acute effects of static, dynamic, and proprioceptive neuromuscular facilitation stretching on muscle power in women. *J Strength Cond Res* 22: 1528–1534, 2008.
63. Masamoto, N, Larson, R, Gates, T, and Faigenbaum, A. Acute effects of plyometric exercise on maximum squat performance in male athletes. *J Strength Cond Res* 17: 68–71, 2003.
64. McCann, MR and Flanagan, SP. The effects of exercise selection and rest interval on postactivation potentiation of vertical jump performance. *J Strength Cond Res* 24: 1285–1291, 2010.
65. McMillian, DJ, Moore, JH, Hatler, BS, and Taylor, DC. Dynamic vs. static-stretching warm up: The effect on power and agility performance. *J Strength Cond Res* 20: 492–499, 2006.
66. Mitchell, CJ and Sale, DG. Enhancement of jump performance after a 5-RM squat is associated with postactivation potentiation. *Eur J Appl Physiol* 13: 1–11, 2011.
67. Needham, RA, Morse, CI, and Degens, H. The acute effect of different warm-up protocols on anaerobic performance in elite youth soccer players. *J Strength Cond Res* 23: 2614–2620, 2009.
68. Pearce, AJ, Kidgell, DJ, Zois, J, and Carlson, JS. Effects of secondary warm up following stretching. *Eur J Appl Physiol* 105: 175–183, 2009.
69. Rassier, DE and MacIntosh, BR. Coexistence of potentiation and fatigue in skeletal muscle. *Braz J Med Biol Res* 33: 499–508, 2000.
70. Rixon, KP, Lamont, HS, and Bembem, MG. Influence of type of muscle contraction, gender, and lifting experience on postactivation potentiation performance. *J Strength Cond Res* 21: 500–505, 2007.
71. Robbins, DW and Docherty, D. Effect of loading on enhancement of power performance over three consecutive trials. *J Strength Cond Res* 19: 898–902, 2005.
72. Rosnow, RL and Rosenthal, R. Statistical procedures and the justification of knowledge in psychological science. *Am Psychol* 44: 1276–1284, 1989.
73. Saez, SV, Gonzalez-Badillo, JJ, and Izquierdo, M. Optimal warm-up stimuli of muscle activation to enhance short and long-term acute jumping performance. *Eur J Appl Physiol* 100: 393–401, 2007.
74. Sale, DG. Postactivation potentiation: Role in human performance. *Exerc Sport Sci Rev* 30: 138–143, 2002.
75. Samuel, MN, Holcomb, WR, Guadagnoli, MA, Rubley, MD, and Wallmann, H. Acute effects of static and ballistic stretching on measures of strength and power. *J Strength Cond Res* 22: 1422–1428, 2008.
76. Scott, SL and Docherty, D. Acute effects of heavy preloading on vertical and horizontal jump performance. *J Strength Cond Res* 18: 201–205, 2004.
77. Sekir, U, Arabaci, R, Akova, B, and Kadagan, SM. Acute effects of static and dynamic stretching on leg flexor and extensor isokinetic strength in elite women athletes. *Scand J Med Sci Sports* 20: 268–281, 2010.
78. Small, K, Mc, NL, and Matthews, M. A systematic review into the efficacy of static stretching as part of a warm-up for the prevention of exercise-related injury. *Res Sports Med* 16: 213–231, 2008.
79. Smilios, I, Pilianidis, T, Sotiropoulos, K, Antonakis, M, and Tokmakidis, SP. Short-term effects of selected exercise and load in contrast training on vertical jump performance. *J Strength Cond Res* 19: 135–139, 2005.

80. Smith, JC and Fry, AC. Effects of a ten-second maximum voluntary contraction on regulatory myosin light-chain phosphorylation and dynamic performance measures. *J Strength Cond Res* 21: 73–76, 2007.
81. Thompson, AG, Kackley, T, Palumbo, MA, and Faigenbaum, AD. Acute effects of different warm-up protocols with and without a weighted vest on jumping performance in athletic women. *J Strength Cond Res* 21: 52–56, 2007.
82. Till, KA and Cooke, C. The effects of postactivation potentiation on sprint and jump performance of male academy soccer players. *J Strength Cond Res* 23: 1960–1967, 2009.
83. Tillin, NA and Bishop, D. Factors modulating post-activation potentiation and its effect on performance of subsequent explosive activities. *Sports Med* 39: 147–166, 2009.
84. Torres, EM, Kraemer, WJ, Vingren, JL, Volek, JS, Hatfield, DL, Spiering, BA, Ho, JY, Fragala, MS, Thomas, GA, Anderson, JM, Häkkinen, K, and Maresh, CM. Effects of stretching on upper body muscular performance. *J Strength Cond Res* 22: 1279–1285, 2008.
85. Turki, O, Chaouachi, A, Behm, DG, Chtara, H, Chtara, M, Bishop, D, Chamari, K, and Amri, M. The effect of warm-ups using different volumes of dynamic stretching on 10-m and 20-m sprint performance in highly trained male athletes. *J Strength Cond Res* 2011.
86. Unick, J, Kieffer, HS, Cheesman, W, and Feeney, A. The acute effects of static and ballistic stretching on vertical jump performance in trained women. *J Strength Cond Res* 19: 206–212, 2005.
87. Vandervoort, AA, Quinlin, J, and McComas, AJ. Twitch potentiation after voluntary contraction. *Exp Neurol* 81: 141–152, 1983.
88. Weber, KR, Brown, LE, Coburn, JW, and Zinder, SM. Acute effects of heavy-load squats on consecutive squat jump performance. *J Strength Cond Res* 22: 726–730, 2008.
89. Weldon, SM and Hill, RH. The efficacy of stretching for prevention of exercise-related injury: A systematic review of the literature. *Man Ther* 8: 141–150, 2003.
90. Woods, K, Bishop, P, and Jones, E. Warm-up and stretching in the prevention of muscular injury. *Sports Med* 37: 1089–1099, 2007.
91. Yamaguchi, T and Ishii, K. Effects of static stretching for 30 seconds and dynamic stretching on leg extension power. *J Strength Cond Res* 19: 677–683, 2005.
92. Yamaguchi, T, Ishii, K, Yamanaka, M, and Yasuda, K. Acute effects of dynamic stretching exercise on power output during concentric dynamic constant external resistance leg extension. *J Strength Cond Res* 21: 1238–1244, 2008.
93. Young, WB, Jenner, A, and Griffiths, K. Acute enhancement of power performance from heavy load squats. *J Strength Cond Res* 12: 82–84, 1998.